

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Mathematical Foundations for Data Science [MAT_5129]

Marks: 50

Duration: 180 mins.

PART-B

Answer all the questions.

- 1) Let X be Bernoulli(p) random variable. The probability mass function of X is (3)

A) $P(X = x) = p^x(1 - p)^{1-x}, x = 0, 1, 0 < p < 1.$
Find the MLE of p .

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) A laboratory measures the weight of $n = 12$ samples of a chemical. The sample mean is 100.5 grams and the sample standard deviation is 0.8 grams. Test at the 5% significance level whether the mean weight significantly differs from the target mean 101 grams. (use the test statistic value=21.920). (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) A random sample of 15 observations gave sample standard deviation 3.10. Test, at the $\alpha = 5\%$ level, whether the population variance equals 9. (use the test statistic value=26.119). (3)

Course Outcome: "CO2" / Cognitive Level: "Remember"

- 2) Setup the ANOVA table for the following per acre production data for 3 varieties of wheat, each grown for 4 fields. State if the different varieties have significant difference in production. (4)
- A)

Field	Variety of wheat		
	A	B	C
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

- B) Customers arrive at a bank according to a Poisson process with rate $\lambda = 4$ per hour. Find the probability that (3)
- no customer arrived in first 30 minutes.
 - atleast 2 customer arrive in first 60 minutes.
 - exactly one customer arrived in first 30 minutes.

- C) A city experiences four types of weather conditions: (3)
- $S_1 = \text{Sunny}, S_2 = \text{Cloudy}, S_3 = \text{Rainy}, S_4 = \text{Stormy}.$
- The following information is known:
- If today is Sunny, tomorrow is Sunny with probability 0.6, Cloudy with 0.3, and Rainy with 0.1.
 - If today is Cloudy, tomorrow is Sunny with probability 0.4, Cloudy with 0.4, Stormy with 0.2.
 - If today is Rainy, tomorrow is Rainy with probability 0.5, Cloudy with 0.3, Stormy with 0.2.
 - If today is Stormy, tomorrow is Rainy with probability 0.6 and Stormy with 0.4.
- Write the transition probability matrix and draw the Markov chain diagram.

- 3) (4)
- A)

Consider a Markov chain with state space $S = \{1,2,3,4\}$. The transition probability matrix is given as:

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0.3 & 0.2 & 0.4 & 0.1 \\ 0.2 & 0.1 & 0.3 & 0.4 \end{bmatrix}$$

- Identify the transient and absorbing states.
- Identify the matrices R and Q from the canonical form.
- Compute the fundamental matrix $N = (I - Q)^{-1}$.
- Find the matrix of absorption probabilities B .

Course Outcome: "CO2" / Cognitive Level: "Understand"

- B) Check whether the states of the Markov chain are recurrent or transient. $S = \{1,2,3,4,5\}$ is the state space and the transition probability matrix is (3)

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{3}{4} \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Define doubly stochastic matrix and write one example of a doubly stochastic matrix with at least three states. Find the stationary distribution of the doubly stochastic matrix that you have written. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Suppose $X \sim N_3(\mu, \Sigma)$. Find the distribution of (3)
- A) $[X_1 - X_2, X_2 - X_3]'$.

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Suppose $X \sim N_3(\mu, \Sigma)$. The mean vector $\mu = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}'$. The covariance matrix $\Sigma = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & 2 \end{bmatrix}$. Find the distribution of $X_2 | (X_1 = 2, X_3 = 0)$. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) (4)

- a) Discuss the Hotelling's T^2 test for the testing of mean of a multivariate population.
b) Write the distribution of T^2 statistic.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Suppose the random variables X_1, X_2 have the covariance matrix $\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. The eigen values of Σ matrix are 3, 1. The corresponding eigen vectors of these eigen values are $\begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}, \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}$, respectively. Calculate
A) a) The two principal components
b) Total variability of the principal components
c) proportion of variability by the first principal component
c) correlation between first principal component and X_2 .

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) The covariance matrix of three random variables X_1, X_2, X_3 is given as $\Sigma = \begin{bmatrix} 5 & 2 & 3 \\ 2 & 3 & 1.5 \\ 3 & 1.5 & 3 \end{bmatrix}$. Assume $m = 1$ factor model . Find
a) Matrix of factor loading (L) for the orthonormal factor model.
b) Matrix of special variances (Ψ).
c) Total communality of X_1, X_2, X_3 .

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Explain the principal component analysis model in details.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025-PG MAT_5129:Mathematical Foundations for Data Science

#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	17
3	3	CO 3	9
4	4	CO 4	7
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG MAT_5129:Mathematical Foundations for Data Science

#	Level	No of Questions	Marks (%)
1	Remember	1	3(6%)
2	Understand	3	10(20%)
3	Application	7	24(48%)
4	Analysis	4	13(26%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)